Mark schemes

Q1.

(a) I > III dominance

E1

1

(b) $v \le 4p_1 + 2p_2$

Ignore any p_3 values

B1 × 3

3

$v \le 2p_1 + 10p_2$

$v \le 6p_1 + p_2$

(c) (i)

p	v	p_1	p_2	r	s	t	u	
1	0	–2	–10	0	0	1	0	0
0	0	1	1	1	0	0	0	1
0	0	–2	8	0	1	–1	0	0
0	1	–2	–10	0	0	1	0	0
0	0	–4	9	0	0	–1	1	0

For row reduction

M1

For correctly having only 1 non-zero in v

A1

Top row correct

A1*

All correct
(*)*OE*

A1*

4

(ii) Negative values

E1

1

(iii) $5P = 18$ $\quad P = 3\frac{3}{5}$

B1

$p_1 = \frac{4}{5}$

B1

$p_2 = \frac{1}{5}$

B1

3

(d) B plays

I	prob	p
II	prob	q
III	prob	$(1 - p - q)$

M1

$\therefore\ 4p + 2q + 6(1 - p - q) = 3\frac{3}{5}$

$2p + 10q + 1 - p - q = 3\frac{3}{5}$

M1

$\Rightarrow 5p + 10q = 6$

$\underline{5p + 45q = 13}$

A1

$\Rightarrow q = \frac{1}{5}$

B1

$p = \frac{4}{5}$

B1

6

[18]

Examiner reports

Q1.

This question combined Game Theory and the Simplex algorithm. Parts (a) and (b) were well answered. Candidates knew the method for part (c)(i) but there were many numerical slips. Although strict use of the Simplex algorithm requires a reduction of the pivot to 1, candidates were not penalised for using integer values in all their working.

Part (d) did not require the Simplex method. Candidates who used the three probabilities p, q and $(1 - p - q)$ for Bharveen were usually able to complete the question successfully.